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House prices and demographic dynamics

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1 Abstract

The main driver of demand for housing in the United States is people. The American population is a diverse group with multifaceted and individualistic housing needs. Housing is a long-lasting investment, so investment decisions made today in the type of housing built will shape the availability of housing long into the future. More specifically, population growth is a key fundamental variable in explaining house price appreciation. “The earth is becoming much more populated, creating more demands for assets,” explains the researcher Dragana Cvijanovic.[1]
“Previous literature focuses on demographics and consumption, but not so much on real estate finance; meanwhile, real estate assets constitute 54% of the world’s wealth.”
Demographics are a particularly useful angle of study for explaining house price appreciation because of the simplicity of the concept.

2 Literature Review

There is an extensive literature linking house prices to demographic dynamics and macroeconomic factors. In the paper ”New in Town: Demographics, Immigration, and the Price of Real Estate”[1] analyses how natural and non-natural components of expected population growth influence house prices. This work shows that house prices are strongly related to the density of population in a certain country. They also observe that non natural component of population growth forecasts cross-sectional variation in house prices. To extend our analyses on house prices we decide to consider also economic-related papers. In the paper ”Measuring the effects of monetary policy on house prices and the economy”[2] by John C Williams the effect of monetary policy in the US between 1976 and 2014 on house prices are studied. According to the study the effects on house prices relative to GDP appear to be robust. It also analysed how inflation effects house prices and it is discovered that inflation moves in the same direction as house prices. In addition, we also consider a paper issued by the ECB (Europen Central Bank) titled ”House Prices, money, credit and the macroeconomy”[13] written by Charles Goodhart and Boris Hofmann. This analyse shows that there is evidence of a significant multidirectional link between house prices and the macroeconomy in European countries between 1970 and 2006. It illustrates how shocks to GDP and the CPI have significant effects on house prices. Obviously, this analyse is made taking into account European countries, but we are going to consider it as well for our analysis based on the United States. Because this is where the financial crisis of 2007 and 2008 started.

3 Introduction

We focus our project on house prices in the United States during the first decade of 2000. First of all, we divide our study’s explanatory variable of demographic change into different basic components: births, deaths, natural increase, net migration, crude birth rate and crude death rate. To be clear, *births* and *deaths* represent the total number of people who were born and died during the year taken into consideration. The variable *natural increase* is computed as difference of the *births* and *deaths* during the year. *Net migration* represents the difference between immigration into and emigration from the area during the year. Therefore, net migration is negative when the number of emigrants exceeds the number of immigrants. The *crude birth rate* indicates the number of live births occurring during the year, per 1,000 population estimated at midyear.

$$CrudeBirthRate = \frac{number \quad of \quad births \quad in \quad 1 \quad year}{thousand \quad total \quad population}$$

Similarly, the *crude death rate* indicates the number of deaths occurring during the year, per 1,000 population estimated at midyear.

$$CrudeDeathRate = \frac{number \quad of \quad deaths \quad in \quad 1 \quad year}{thousand \quad total \quad population}$$

Remark: by subtracting the crude death rate from the crude birth rate provides the rate of natural increase, which is equal to the rate of population change in the absence of migration.

In terms of the so called "non-natural" component of demographic change (immigration), an essential factor to take into account is that market shocks drive people to immigrate en masse from lesser to hotter economic areas. The key difference between the natural (birth and death) and non-natural components is their degree of predictability. Population growth through natural causes follows an expected course, but, due to its sudden and market-driven nature, population growth via immigration does not. We ultimately find that population growth drives house price appreciation, but also that this effect is predominantly driven by the non-natural component rather than the natural. This effect can be attributed to the unpredictability of immigration. "Builders can anticipate a fraction of population growth, the fraction that is due to natural causes (birth and death), but the portion that is due to immigration cannot be predicted, which is at least partly why immigration drives house appreciation — because of its unanticipated shock to the market." In other words, with natural growth, builders react accordingly to market demands: they offset anticipatable demand by providing more supply. But they can only slightly predict growth via immigration, so they underreact to that demand, which drives up house prices. Furthermore, to explain the house prices trend to a deeper level, we have extended our model by adding two new predictors: CPI and GDP. CPI stands for "Consumer Price Index". and it is a complete measure which is exploited in order to estimate price changes in a basket of goods and services representative of consumption expenditure in an economy. The consumer price index is computed by national statistical agencies based on several indices. More precisely, various categories and sub-categories have been made for classifying consumption items and on the basis of consumer categories like urban or rural in order to calculate the CPI. The consumer price index can be considered one of the most important statistics for an economy and is generally based on the weighted average of the prices of commodities. As a matter of fact, it gives an idea of the cost of living. In addition, the CPI is usually used to measure inflation. The percentage change in this index over a period of time gives the amount of inflation over that specific period, i.e. the increase in prices of a representative basket of goods consumed.

GDP stands for "Gross Domestic Product" and it represents the total monetary or market value of all the finished goods and services produced within a country's borders in a specific time frame. As a broad measure of overall domestic production, it functions as a comprehensive score of a given country's economic health. For instance, in the United States, the government releases an annualized GDP estimate for the calendar year. The individual data sets included in this report are given in real terms, so the data is adjusted for price changes and is, therefore, net of inflation.

4 Data

Our window of analysis is from 2000 and 2011, also known as "The aughts". As a very first step we make some plot in order to visualize the setting. We take into account the population mid-year estimates total and we consider separately females and males over the time period.

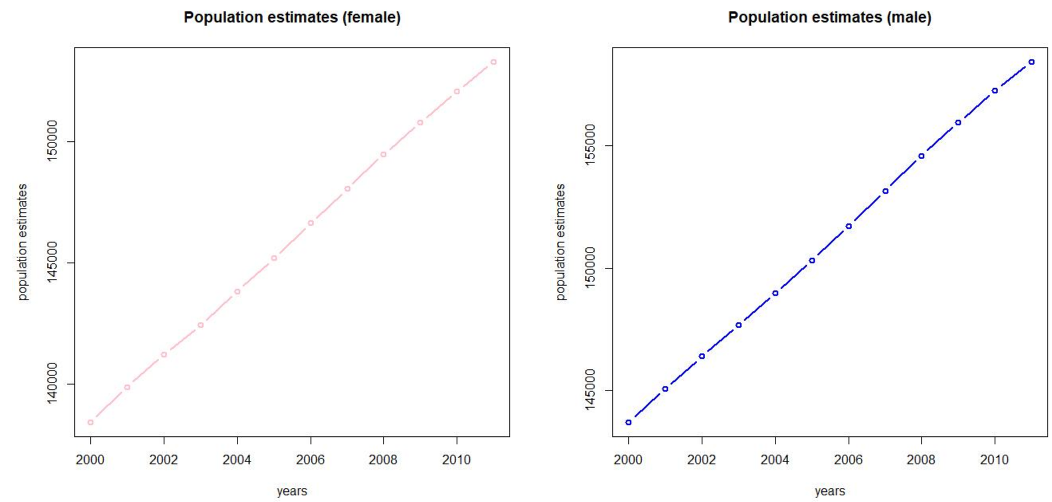

Figure 1: Trend of the female and male population in the US over the years

The graphs show that both the females and males mid-year estimates total have a positive linear trajectory over the time. Hence, it can be stated that during the first decade of 2000 the overall population has risen steadily. On the other hand, as this last image illustrates the population estimates growth rate has collapsed between 2000 and 2011.

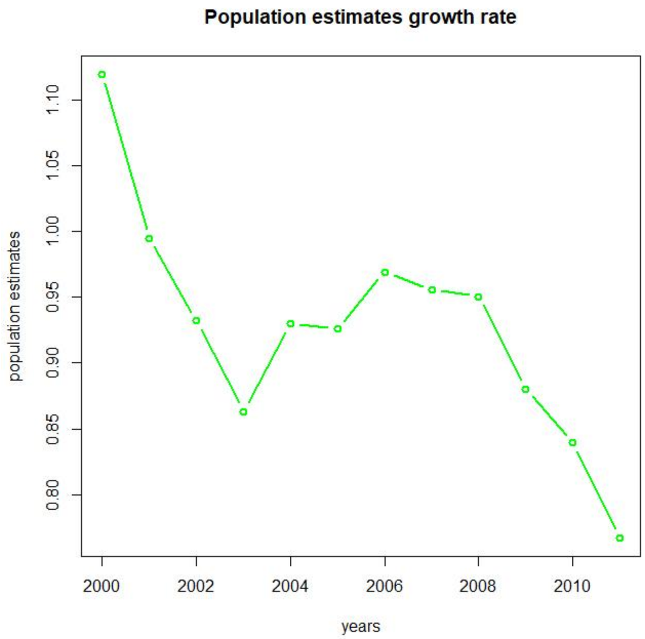

Figure 2: Population estimates growth rate trend over the years

In figure 3, house prices rate over the years is shown. As the image illustrates, the rate dramatically decreases between 2006 and 2008 which can be explained by the "Financial crisis of 2007–2008".
As a matter of fact, the crisis was the result of cheap credit and lax lending standards that fueled a housing bubble. The financial crisis of 2007-2008 was years in the making. By the summer of 2007, financial markets around the world were showing signs that the reckoning was overdue for a years-long binge on cheap credit. Two Bear Stearns hedge funds had collapsed, BNP Paribas was warning investors that they might not be able to withdraw money from three of its funds, and the British bank Northern Rock was about to seek emergency funding from the Bank of England. Yet despite the warning signs, few investors suspected that the worst crisis in nearly eight decades was about to engulf the global financial system, bringing Wall Street's giants to their knees and triggering the Great Recession. It was an epic financial and economic collapse that cost many ordinary people their jobs, their life savings, their homes, or all three.[2].

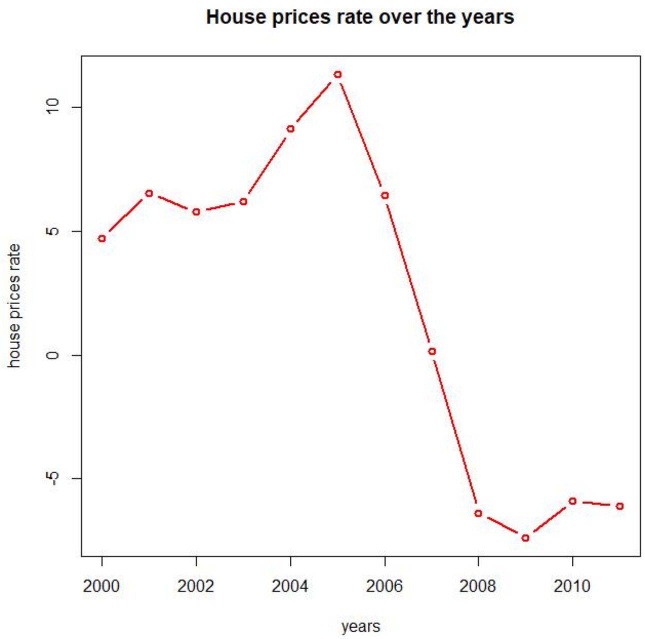

Figure 3: The image shows the house prices trend over the years

Afterwards, in order to investigate the Structural Change, we perform an **sctest**. The "**sctest**" is a convenience interface for performing structural change tests in linear regression models. Given a model it is tested whether the data support the null hypothesis that there is a stable structure against the alternative that it changes over time. As we can see from the trend of house prices over the years in the US, there has been a significant fall around 2005/2008, and this is the main reason why we needed to perform this kind of test. Structural change is of central interest in many fields of research and data analysis: to learn if, when and how the structure of the data generating mechanism underlying a set of observations changes. Usually, it is known with respect to which quantity the structural change might occur, e.g., over time or with the increase of a certain risk factor. One of the simplest examples for such a structural change is a time series whose mean changes at a single breakpoint.
We obtain the following ouputs:

```
          supF test

data:  fs.y
sup.F = 65.725, p-value = 3.662e-14
```

Figure 4

```
aveF test

data:  fs.y
ave.F = 18.674, p-value < 2.2e-16
```

Figure 5: The image shows the House prices rate over the years

The output shows a significant small p-value, both with the F statistics and the asymptotic χ^2. This means that there is statistical evidence to reject the null hypothesis and confirm the occurrence of such change in house prices in the USA.

At the same time we investigate other variables that affected the population. In figure 6 it is illustrated the trend of net migration in USA over the 11 years. As the image shows the net migration has faced a significant decrease. This result is not unexpected. As a matter of fact, after 9/11 terrorist attacks to the Twin Towers in New York City the US government hardened its migration policies. In addition, in 2007-2008, the overall U.S. migration rate reached its lowest point since World War II. The slowdown was especially pronounced for long-distance moves, which remained flat through 2008–2009, as well as for single people and renters. Both long-distance and short-distance movers were less likely to cite housing reasons for their moves.

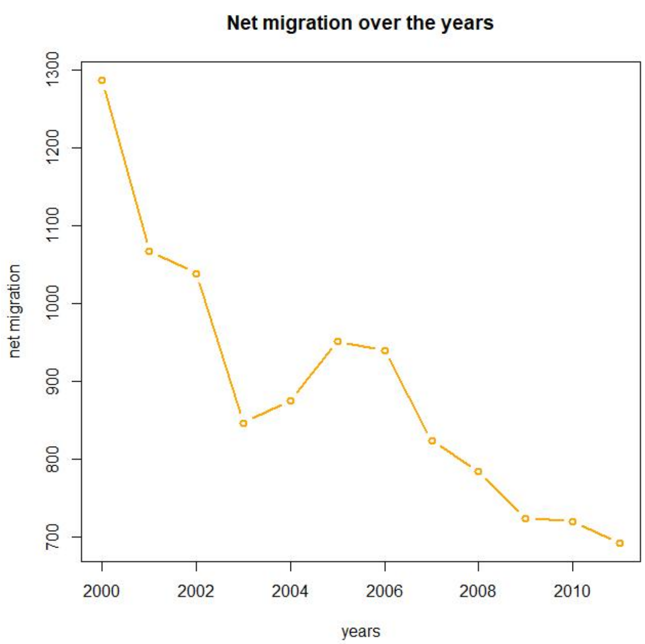

Figure 6: The image shows the Total Net migration the years

5 Methodology and Models with Results

5.1 Linear Regression for the dependent variable "Population growth"

We start our data analysis with a linear regression model based on different population independent variables: *birth*, *death*, *net migration*, *crude_birth*, *crude_death* . We obtain the following result:

Table 1

	Dependent variable:
	y
birth	44.117
	(26.337)
death	47.804
	(43.625)
net_migration	1.673**
	(0.641)
crude_birth	−14,063.660
	(7,952.138)
crude_death	−14,489.010
	(13,170.580)
Constant	311,346.900***
	(4,005.116)
Observations	12
R^2	1.000
Adjusted R^2	1.000
Residual Std. Error	156.218 (df = 6)
F Statistic	8,643.510*** (df = 5; 6)
Note:	*p<0.1; **p<0.05; ***p<0.01

The regression model produces a high R^2 value (≈ 1), and a low global p-value ($\approx 10^{-11}$).
This combination indicates that changes in the predictors are related to changes in the response variable and that the model explains a lot of the response variability. The most statistically significant predictor variable is the *"Net migration"* variable with a **p-value = 0.04**.
We can conclude that the population growth is mostly influenced by the difference between immigration and emigration of different geographic areas. On the other hand, we can see that the other regressors don't seem to be significant as they have a high p-value.

In order to improve the model we try to see what happens if we fit the linear regression with just the *"Net migration"* predictor. Before doing that, we have to test the linear hypothesis that all other regressors can be equal to 0, that is they are not significant in explaining our model.

Table 2

Statistic	N	Mean	St. Dev.	Min	Max
Res.Df	2	8.000	2.828	6	10
RSS	2	102,749,047.000	145,102,021.000	146,424.200	205,351,670.000
Df	1	4.000		4	4
Sum of Sq	1	205,205,245.000		205,205,245.000	205,205,245.000
F	1	2,102.165		2,102.165	2,102.165
Pr(>F)	1	0.000		0	0

The p-value of the test is very low, so we can reject the null hypothesis and we cannot remove all the predictors but *"Net migration"*. Therefore, we perform another test of the linear hypothesis which shows:

Table 3

Statistic	N	Mean	St. Dev.	Min	Max
Res.Df	2	7.000	1.414	6	8
RSS	2	161,398.700	21,177.100	146,424.200	176,373.200
Df	1	2.000		2	2
Sum of Sq	1	29,948.950		29,948.950	29,948.950
F	1	0.614		0.614	0.614
Pr(>F)	1	0.572		0.572	0.572

Given the fact that now the p-value is high, we cannot reject H0 and so we remove from the model the variables *"death"* and *"crude_death"* that are not significant.
As a consequence, we re-estimate the model without these two variables. This procedure leads us to a statistical significant model (*Table 4*): all the predictors show a low p-value, also the global p-value is even lower than before ($\approx 10^{-15}$), and the R^2 value is still pretty high (≈ 1).
In conclusion, we can infer that this is the best model that explains that the growth of the population of the USA in the considered time frame mainly depends on the migration flow and the births.

Table 4

	Dependent variable:
	y
birth	73.053***
	(0.825)
net_migration	1.773**
	(0.603)
crude_birth	−22,798.540***
	(240.684)
Constant	310,816.200***
	(2,034.724)
Observations	12
R^2	1.000
Adjusted R^2	1.000
Residual Std. Error	148.481 (df = 8)
F Statistic	15,945.780*** (df = 3; 8)
Note:	*p<0.1; **p<0.05; ***p<0.01

Therefore, we decide to calculate the confidence intervals (CI). We can notice that all of the intervals of the predictors' coefficients and that of the intercept α do not contain the value 0. This is a further proof of the fact that we can reject the null hypothesis (α=0 and β_i=0), and state that all of these parameters influence the value of y, which is the population. Moreover, if α and the variables *birth* and *net migration* increase, so does the population. On the other hand, if the *crude birth* variable rises, the population decreases.

Table 5

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
2.5 %	3	92,857.630	173,374.000	−14,247.270	−7,156.111	146,410.100	292,885.100
97.5 %	3	175,364.700	300,413.500	−20.733	1,924.550	263,057.400	522,245.000

5.2 Linear Regression for the dependent variable "House Prices"

We continue our data analysis with a linear regression model based on different independent variables: *crude_birth*, *crude_death*, *CPI rate* and *GDP rate*, while *house prices* is the dependent variable.

Table 6

	Dependent variable:
	house_prices
crude_death	-3.425
	(15.379)
crude_birth	5.498
	(4.576)
cpi	0.957
	(1.421)
gdp	-0.00000
	(0.00000)
Constant	-19.049
	(154.338)
Observations	12
R^2	0.635
Adjusted R^2	0.427
Residual Std. Error	5.128 (df = 7)
F Statistic	3.050^{*} (df = 4; 7)
Note:	*p<0.1; **p<0.05; ***p<0.01

As the table shows, all the predictors present an importantly high p-value, between 26,9% and 83%. In addition, this model explains a bit more than the half of the total variability. The global p-value is above the 5% treshold. In general, we can say that this model needs some adjustments.
Since we are seeking an upgrade, we decide to repeat another linear regression (*Table 7*). After implementing a suitable test of the linear hypothesis, this time we add the *"population growth"* variable and remove the *"cpi"* one, and we do not consider the intercept since in the previous model (*Table 6*) it had a large p-value. Finally, we find two significant variables, *"gdp"* and *"crude-birth"*, but *"crude_death"*'s and *"population growth"*'s p-values do not make any improvement. That is why we decide to get rid of them, obtaining a better model (*Table 8*). In conclusion, we can deduce that the house prices during this decade is mainly influenced by the Gross Domestic Product of the population and the crude birth rate.

Table 7

	Dependent variable:
	house_prices
population_growth	−39.463
	(26.751)
crude_death	−9.864
	(6.296)
crude_birth	11.837*
	(5.207)
gdp	−0.00000***
	(0.00000)
Observations	12
R^2	0.722
Adjusted R^2	0.582
Residual Std. Error	4.396 (df = 8)
F Statistic	5.182** (df = 4; 8)
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 8

	Dependent variable:
	house_prices
crude_birth	2.471***
	(0.655)
gdp	−0.00000***
	(0.00000)
Observations	12
R^2	0.606
Adjusted R^2	0.527
Residual Std. Error	4.678 (df = 10)
F Statistic	7.681*** (df = 2; 10)
Note:	*p<0.1; **p<0.05; ***p<0.01

We calculate the confidence intervals for the last model. Both of the variables' intervals do not contain the 0: we can then reject the H0, under which β_i=0 (i=1, 2). This result is predictable and coherent since the p-values were all under the 5%.

Table 9

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
2.5 %	2	0.506	0.716	−0.00000	0.253	0.759	1.013
97.5 %	2	1.965	2.779	−0.00000	0.983	2.948	3.930

We create a scatterplot, comparing house prices with respectively Population mid-year estimates, Births, Deaths and Net migration.

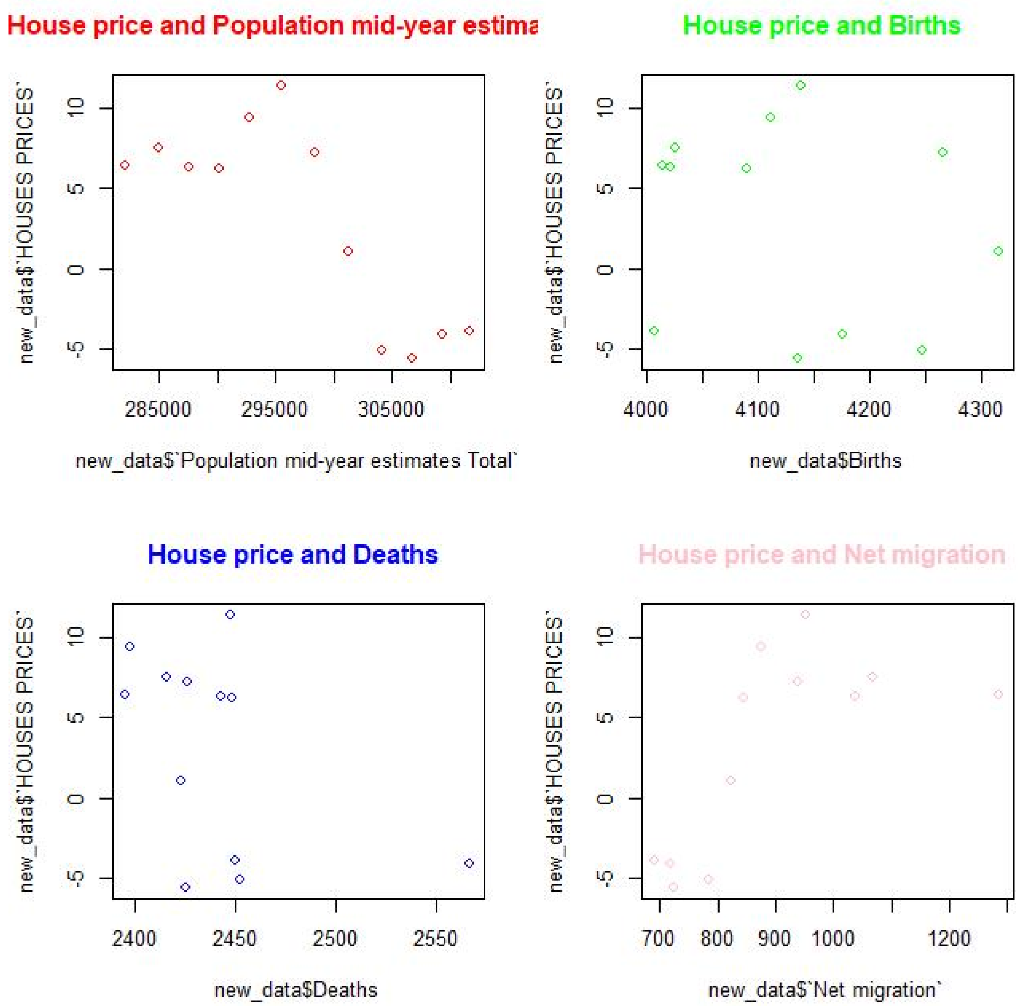

Figure 7

From the graphs we can deduce that between house prices and births and between house prices and deaths there is not any trend. Then, we plot the residuals for the model on *Table 4* and we can notice that in the graph the red line representing the mean is not constant to zero. However, we can not reject the homoscedasticity assumption since we do not have so many data.

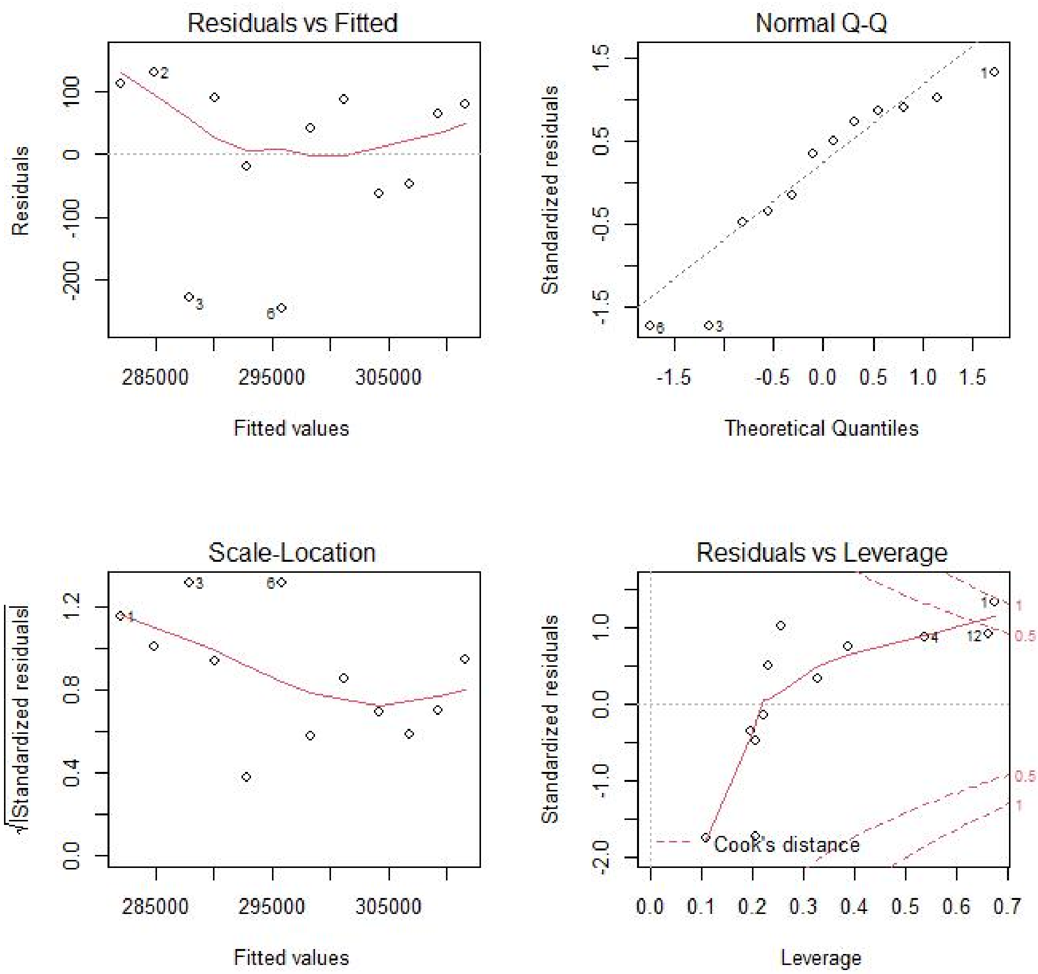

Figure 8

Similarly, we plot the residual for the model on *Table 7*. As before, it can be noticed that in the graph the red line is not constant to zero. However, we can not reject the homoscedasticity hypothesis since we do not have many data.

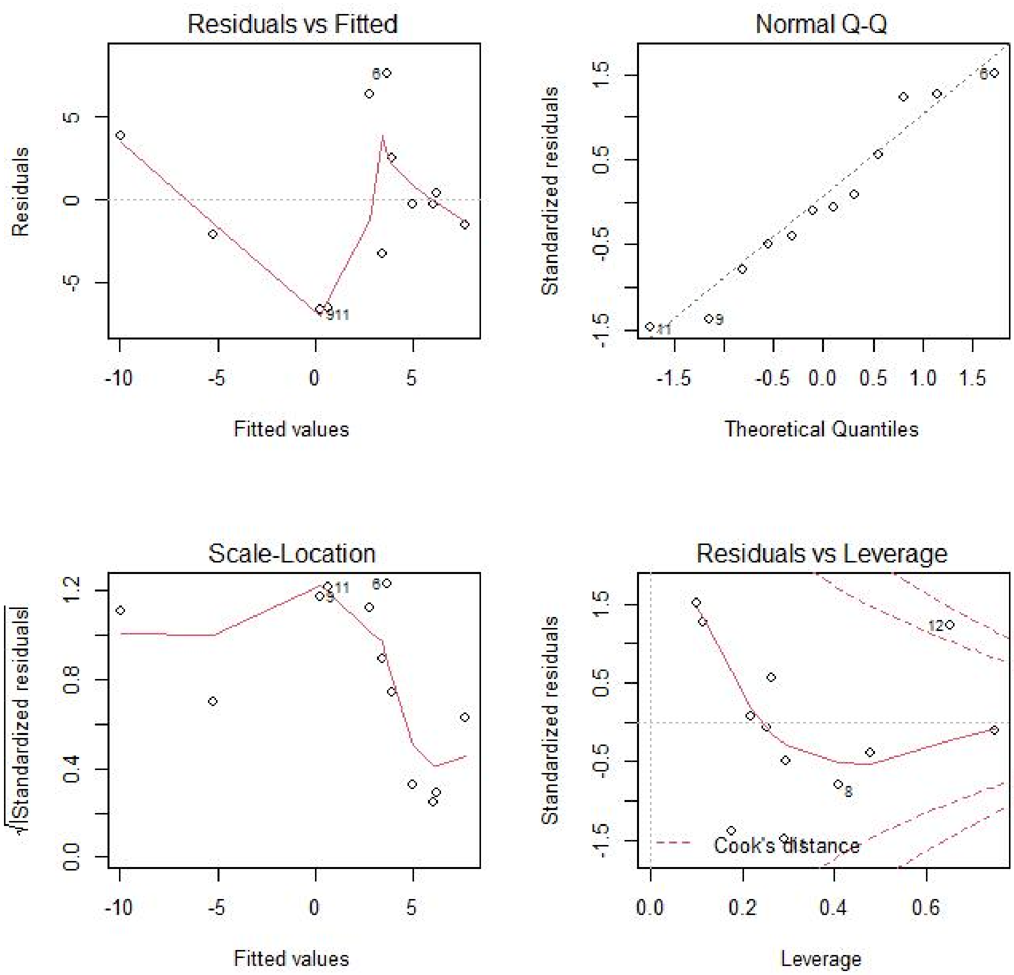

Figure 9

Since from the assumptions we have an homoscedastic model we do not implement the WLS test, which is used to fix heteroskedasticity. In order to prove that we cannot reject the homoscedasticity assumption we decide to exploit the White test and Breusch-Pagan test for the model in *Table 4*. From these tests we obtain the following p-value 0.531 and 0.5991, respectively.

```
	studentized Breusch-Pagan test

data:  fit2
BP = 2.2048, df = 3, p-value = 0.531
```

Figure 10

```
	Breusch-Pagan test

data:  fit2
BP = 1.8732, df = 3, p-value = 0.5991
```

Figure 11

Since the p-values are high we cannot reject the null hypothesis. We can conclude that the homoscedasticity assumption cannot be declined.

Then, we do the same two tests for the model in *Table 7*.

```
	studentized Breusch-Pagan test

data:  fit4
BP = 0.65508, df = 3, p-value = 0.8837
```

Figure 12

```
	Breusch-Pagan test

data:  fit4
BP = 0.89725, df = 3, p-value = 0.8261
```

Figure 13

We can notice that also for this model we obtain high p-values. Therefore, we can conclude that the homoscedasticity assumption cannot be declined.

5.3 Autoregressive Model of the first order for the dependent variable "Population growth"

At this point, we decide to implement the autoregressive model of the first order for the dependent variable *Population growth*. As *Table 10* shows: *birth_1* and *crude birth_1* are not significant. This means that birth and crude birth of the previous year and net migration of the current year do not influence on the value of the growth of the population of the current year. On the contrary, *net_migration_1*, that is the value of the migration flow of the previous year, has a 10% significance level. Therefore, it seems to influence the value of the growth of the population of the current year. To sum up, in this AR(1) model also the *net migration* of the current year becomes not significant.

Table 10

	Dependent variable:
	y
birth	67.075*
	(24.442)
net_migration	0.574
	(0.751)
crude_birth	−20,794.660*
	(7,521.343)
birth_1	6.539
	(24.814)
net_migration_1	1.261*
	(0.456)
crude_birth_1	−2,219.766
	(7,384.549)
Constant	311,627.900***
	(3,633.473)
Observations	11
R^2	1.000
Adjusted R^2	1.000
Residual Std. Error	103.139 (df = 4)
F Statistic	12,732.030*** (df = 6; 4)
Note:	*p<0.1; **p<0.05; ***p<0.01

5.4 Autoregressive Model of the second order for the dependent variable "Population growth"

Then, we consider an Autoregressive model of the second order. From the following table we can notice that *birth_2*,*net migration_2* and *crude birth_2* are not significant as they show an high p-value.
This means that birth, net migration and crude birth of two previous year do not influence on the value of the growth of the population of the current year.

Table 11

	Dependent variable:
	y
birth	59.103^{**}
	(18.443)
net_migration	0.708
	(0.681)
crude_birth	$-18{,}239.080^{**}$
	(5,730.954)
L(birth, 2)	14.350
	(19.201)
L(net_migration, 2)	0.521
	(0.607)
L(crude_birth, 2)	$-4{,}550.261$
	(5,644.363)
Constant	$310{,}520.200^{***}$
	(4,262.989)
Observations	10
R^2	1.000
Adjusted R^2	1.000
Residual Std. Error	88.515 (df = 3)
F Statistic	$13{,}056.330^{***}$ (df = 6; 3)
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

5.5 Autoregressive Model of the first order for the dependent variable "House Prices"

At this point, we decide to implement the autoregressive model of the first order for the dependent variable *House Prices*. From *Table 12* we can deduce that no variables influence this year house prices.

Table 12

	Dependent variable:
	house_prices
population_growth	65.375
	(68.740)
crude_death	9.552
	(11.887)
crude_birth	4.163
	(9.864)
population_growth_1	−43.550
	(38.185)
crude_death_1	18.612
	(8.991)
crude_birth_1	−6.369
	(12.994)
Constant	−217.180
	(147.593)
Observations	11
R^2	0.820
Adjusted R^2	0.549
Residual Std. Error	4.733 (df = 4)
F Statistic	3.032 (df = 6; 4)
Note:	*p<0.1; **p<0.05; ***p<0.01

Subsequently, we decide to exploit the *Durbin Watson* test. The result shows a high p-value. Therefore, we can not reject the null hypothesis. Hence we can not exclude the possibility that there is not any autocorellation for the model illustrated in *Table 10*.

```
        Durbin-Watson test

data:  fit2_1
DW = 2.6851, p-value = 0.4279
alternative hypothesis: true autocorrelation is greater than 0
```

As far as the model in *Table 12* is concerned the result is very similar.

```
        Durbin-Watson test

data:  fit4_1
DW = 2.5215, p-value = 0.4587
alternative hypothesis: true autocorrelation is greater than 0
```

In addition, we implement the Breusch-Godfrey test for both the models in *Table 10* and *Table 12*:

```
        Breusch-Godfrey test for serial correlation of order up to 1

data:  fit2_1
LM test = 4.9689, df = 1, p-value = 0.02581
```

```
        Breusch-Godfrey test for serial correlation of order up to 1

data:  fit4_1
LM test = 2.5501, df = 1, p-value = 0.1103
```

The result of the Breusch-Pagan test shows that there is not any autocorrelation for both models.

Furthermore, we do also the *Box_Ljung* and *Box_pierce* tests for both models. We obtain the following results:

```
        Box-Ljung test

data:  fit2_1$residuals
X-squared = 5.4409, df = 6, p-value = 0.4886


        Box-Pierce test

data:  fit2_1$residuals
X-squared = 3.7786, df = 6, p-value = 0.7066


        Box-Ljung test

data:  fit4_1$residuals
X-squared = 7.5201, df = 6, p-value = 0.2754


        Box-Pierce test

data:  fit4_1$residuals
X-squared = 4.5399, df = 6, p-value = 0.604
```

These results illustrate that all p-values are high. Hence, both tests confirm the result previously obtained with different tests, that there is not any autocorrelation.
Then, we plot the Autocorrelation Functions:

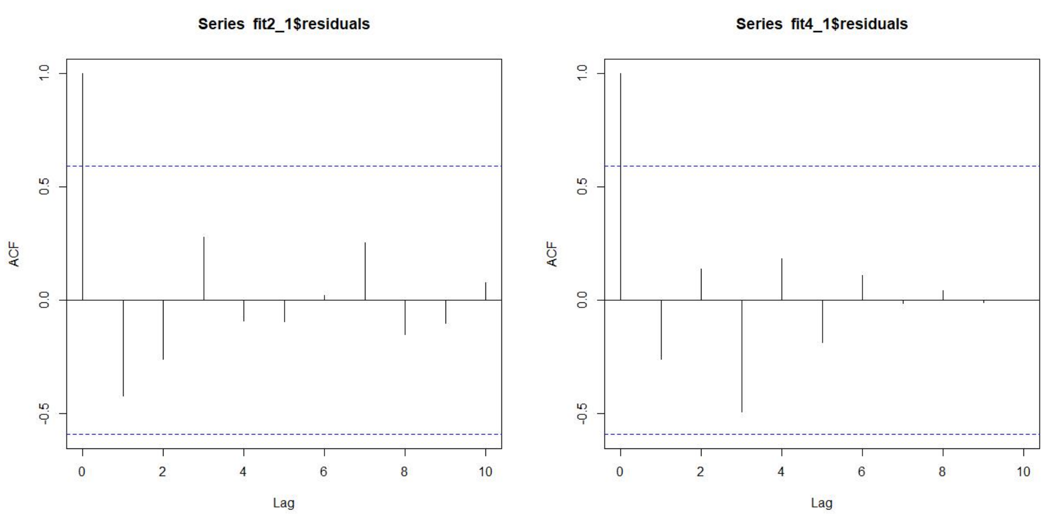

Figure 14: Trend of the female and male population in the US over the years

The graphs show that there is not any autocorrelation in the residuals. The first vertical line has value 1, because it represents the autocorrelation with itself. Then, there is the autocorrelation between the respective lag and today. In conclusion we can say that it is a good model because in the residuals nothing is remaining.

5.6 Time series

First of all, time series data are data collected on the same observational unit at multiple time periods. A time series can also be defined as a collection of random variables Xt. Such a collection of random variables ordered in time is called stochastic process. We decide to implement the time series of the *House Prices* variable with respect to the *cpi* (Consumer Price Index) up to the fourth lag. The result are shown in the following table:

Table 13

	Dependent variable:
	d(house_prices)
cpi	−0.741
	(0.409)
L(cpi, 1)	−1.153*
	(0.343)
L(cpi, 2)	−1.619**
	(0.361)
L(cpi, 3)	−3.907**
	(0.678)
L(cpi, 4)	3.482*
	(0.941)
Constant	8.494
	(4.814)
Observations	8
R^2	0.978
Adjusted R^2	0.923
Residual Std. Error	1.065 (df = 2)
F Statistic	17.857* (df = 5; 2)
Note:	*p<0.1; **p<0.05; ***p<0.01

5.7 Model Selection Criteria

Since we have to compare several models with different sets of parameters, we want to know how well the models explain the data and also which model explains the data better. What we are looking for is a criterion based on which we can choose the most suitable model available. It can be suggested that increasing the number of parameters will reduce the sum of squares of the residuals (the R2, and hence the proportion of total variability explained by the model) since the model uses more data. However, we cannot completely rely only on that, but we need to take into consideration a model complexity. It is possible to use criteria that take into account the complexity of the models and penalize for complexity, thus preferring the more parsimonious ones. The two most commonly used model selection criteria are the Akaike Information Criterion (AIC) and the Bayesian information criteria (BIC).

$$AIC = -ln(L(\theta)) + 2 \cdot k$$

$$BIC = -ln(L(\theta)) + k \cdot ln(n)$$

where

$$k = number \quad of \quad parameters$$

$$n = number \quad of \quad used \quad observations$$

$$L(\theta) = maximum \quad likelihood \quad estimate \quad function$$

Table 14

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
df	5	5.000	1.581	3	4	6	7
BIC	5	51.853	15.307	27.178	47.683	63.651	64.016

Table 15

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
df	5	5.000	1.581	3	4	6	7
AIC	5	50.646	14.946	26.621	46.500	62.424	62.457

The two criteria AIC and BIC are supposed to be as small as possible. If we let the fit of the model grow both criteria will tend to $-\infty$. Both AIC and BIC criteria select from several models the one which is more likely to be the best. It is important to remark that AIC scores are relevant only if they are compared with different AIC scores, coming from the same dataset. The two criteria consist on evaluating the fit of the model on the training data and a penalty term is added for the model's complexity. (It can be compared to Lasso/Ridge regularization in regression theory). As a rule, the lower the value of the AIC/BIC the better it is.

5.8 ARCH Model

Autoregressive conditional heteroskedasticity (ARCH) is a statistical model used to analyze volatility in time series in order to forecast future volatility. In the financial world, ARCH modeling is used to estimate risk by providing a model of volatility that more closely resembles real markets. ARCH modeling shows that periods of high volatility are followed by more high volatility and periods of low volatility are followed by more low volatility. In order to verify if we can implement ARCH model, we decide to calculate the Lagrange Multiplier test (LM test) which tells us if we are in the case of heteroskedasticity:

```
	ARCH LM-test; Null hypothesis: no ARCH effects

data:  d_hp
Chi-squared = 1.1896, df = 1, p-value = 0.2754
```

This result shows that there is no ARCH effects since the p-value is high and we cannot reject the null hypothesis. As a matter of fact, this response is not unexpected since we have previously seen that we have homoscedasticity.

5.9 Principal Component Analysis - PCA

The Principal Components Analysis, also known as PCA, is a technique which is mostly implemented in order to reduce the dimensionality of data while preserving as much as possible the information in the original data. The PCA consists on projecting data onto a lower-dimensional subspace that retains most of the variance among the data points. The idea is that each of the n observations lives in p-dimensional space, however not all of these dimensions are equally interesting. PCA seeks a small number of dimensions that are as interesting as possible, where the concept of interesting is measured by the amount that the observations vary along each dimension. Each of the dimensions found by PCA is a linear combination of the p features. The first principal component of a set of features X_1,X_2, . . . , X_p is the normalized linear combination of the features

$$Z1 = \phi_{11} \cdot X_1 + \phi_{21} \cdot X_2 + ... + \phi_{p1} \cdot X_p$$

that has the largest variance. By normalized, we mean that

$$\sum_{j=1}^{p} \phi_{j1}^2 = 1$$

We refer to the elements ϕ_{11}, . . . , ϕ_{p1} as the loadings of the first principal component; together, the loadings make up the principal component loading vector, $\phi_1 = (\phi_{11}\ \phi_{21}\ .\ .\ .\ \phi_{p1})$' . We constrain the loadings so that their sum of squares is equal to one, since otherwise setting these elements to be arbitrarily large in absolute value could result in an arbitrarily large variance. Given a n $\times$ p data set X, how do we compute the first principal component? Since we are only interested in variance, we assume that each of the variables in X has been centered to have mean zero (that is, the column means of X are zero). We then look for the linear combination of the sample feature values of the form $z_{i1} = \phi_{11}{\cdot}\mathrm{x}_{i1} + \phi_{21}{\cdot}\mathrm{x}_{i2} + . . . + \phi_{p1}{\cdot}\mathrm{x}_{ip}$ that has largest sample variance, subject to the constraint that

$$\sum_{j=1}^{p} \phi_{j1}^2 = 1$$

In other words, the first principal component loading vector solves the optimization problem

$$maximize_{\phi_{11},..\phi_{p1}} \left\{ \frac{1}{n} \cdot \sum_{i=1}^{n} \left(\sum_{j=1}^{p} \phi_{j1} x_{ij} \right)^2 \right\}$$

subject to

$$\sum_{j=1}^{p} \phi_{j1}^2 = 1$$

After the first principal component $Z1$ of the features has been determined, we can find the second principal component $Z2$. The second principal component is the linear combination of X_1,X_2, . . . , X_p that has maximal variance out of all linear combinations that are uncorrelated with Z1. And proceed accordingly. At this point, we try a dimensionality reduction by computing a Principal Component Analysis. From the results of *Explained Variance* it is evident how the first two principal components explain almost all of the total variability.

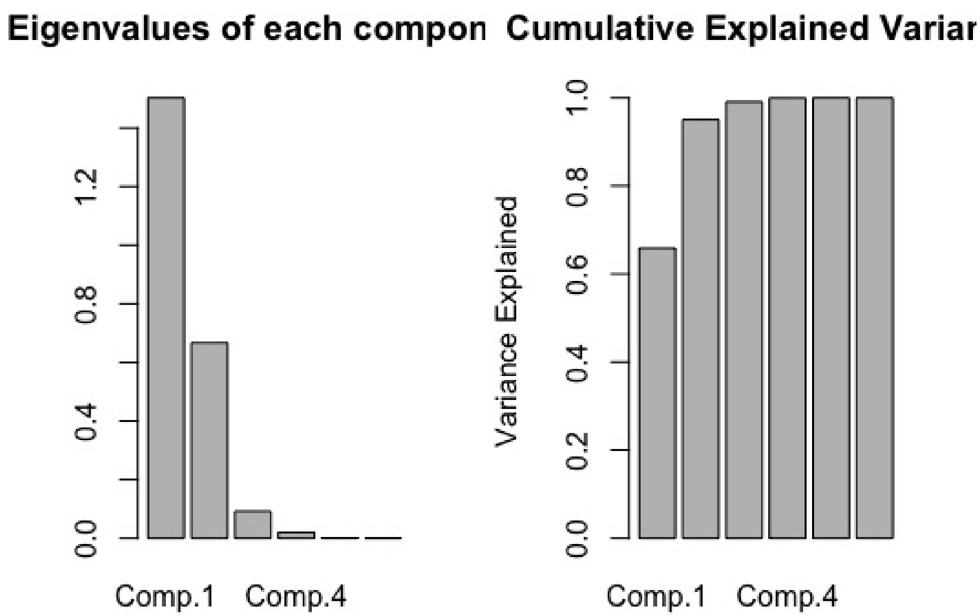

Figure 15: The bar chart on the left is the squared standard deviation, while the one on the right is the Cumulative Explained Variance

As *Table 16* shows, the first component explains the 65,85% of the total variability while the second one the 29,21%.

	$\|Comp.1\|$	$\|Comp.2\|$	$\|Comp.3\|$	$\|Comp.4\|$	$\|Comp.5\|$	$\|Comp.6\|$
Standard Deviation	1.2258375	0.8165688	0.30264038	0.141040900	0.0318712366	1.227600e-06
Proportion of Variance	0.6585025	0.2921980	0.04013705	0.008717296	0.0004451327	6.603976e-13
Cumulative Proportion	0.6585025	0.9507005	0.99083757	0.999554867	1.0000000000	1.000000e+00

Table 16

Coherentely with what we have just observed, the graph of the *contribution to the total variance* (*Figure 16*) shows an "elbow" in correspondence of the second principal component. This means that we can reduce the dimensionality of the data set, considering a smaller one, that contains only the first 2 components, minimizing information loss.

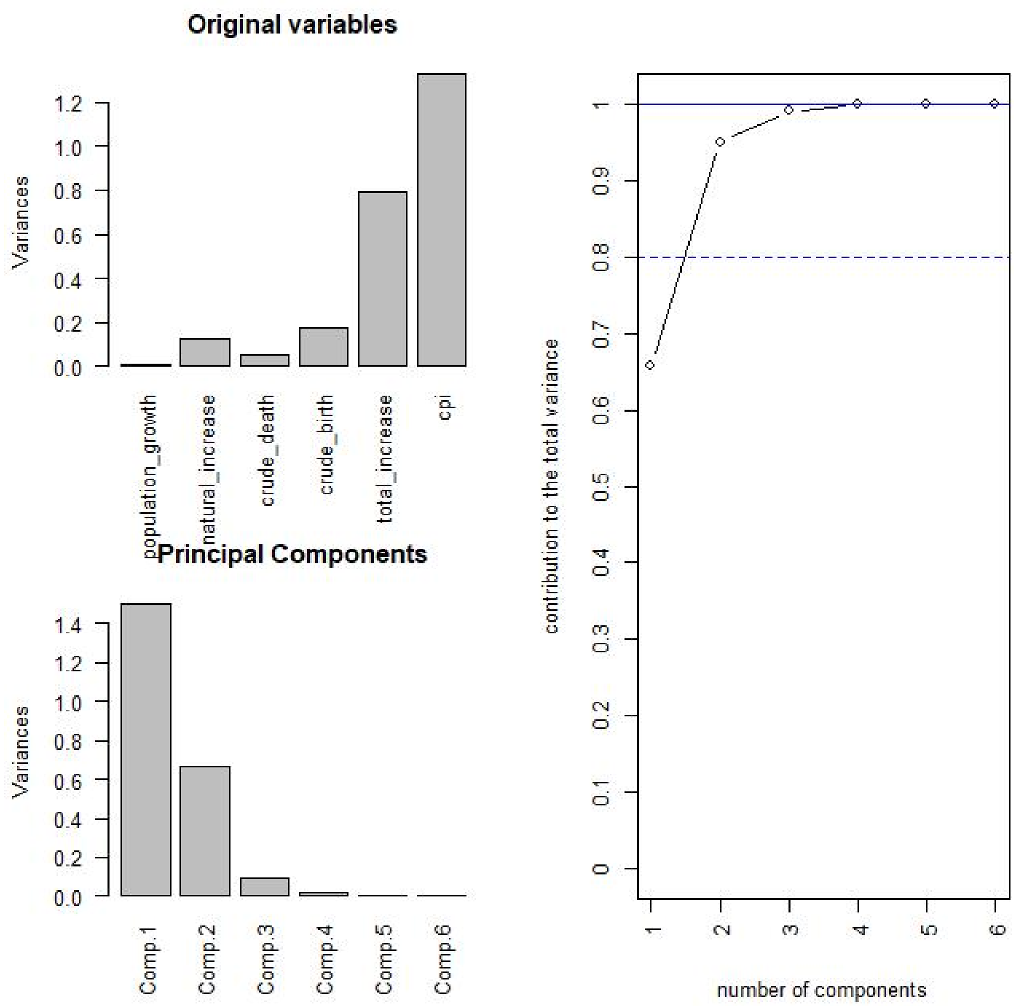

Figure 16: Two bar charts that represents the Original variables and Principal Components, respectively. The line graph on the right represents the contribution to the total variance

6 Conclusion

The purpose of our project was to analyse what are the main factors that influenced house prices in a precise time period (2000-2011). We decided to use both demographic and economic data in order to implement a wider study. The reason why we selected the 2000-2011 is the fact that during this period the US faced terrorist attacks which led to stricter migration policies and a financial crisis which was the result of the housing bubble and the subsequent subprime mortgage crisis.
According to our analysis the GDP rate and the total number of births are the values that affect the prices the most. For instance, if we consider the time period before the financial crisis (2000-2005) in which we can observe only a population growth, we notice that the house prices rate increased significantly. Therefore, we can deduce that house prices increase in value when the population in a local area grows faster than the rate of properties being built. As a result, prices will go up. It's a simple question of supply and demand.
On the contrary, as observed, the financial crisis led to a dramatic decrease of the house prices. Hence, we can infer that during a recession, when there are usually less buyers, houses stay unsold on the market longer and that leads to a decrease in house prices.
In conclusion, we can state that in the economy world, house price changes can have the wealth effect and the collateral effect simultaneously. The wealth effect and the collateral effect respectively refer to changes in desired consumption and changes in actual consumption due to house price variation. While both effects predict causation from house price changes to economic growth, they affect the economy through different channels and have distinctive policy implications. For instance, if an economic decline is caused by the wealth effect instead of the collateral effect of decreasing house prices, which means that households reduce their desired consumption not because they are more financially constrained but because they feel poorer, easing the credit availability may not help stimulate the economy.

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